Towards 2020 Science: A Draft Roadmap

Goals

- LHC preparation
- Systems biology emerges as new discipline
- LHC switch-on
- *Ex vivo* molecular-computer diagnosis
- Automated remote species-identification
- Reliable global warming, natural disaster and weather-prediction models
- Keystone-species identification
- Systems approach to biology becomes mainstream
- *In vivo* molecular-computer diagnosis
- Comprehensive biological data integration: majority of 'omes co-analysable
- Individualised medicine
- Modelling based vaccines
- Finding the Higgs boson?
- Predictive models of: effects of rainforest destruction, forest sustainability, effects of climate change on ecosystems, effects of climate change on foodwebs, restoration ecology planning, global health trends, sustainable agriculture solutions
- Full model of a single cell
- Foundational theory of Global Ecosystems
- Full model of a multi-cellular tissue/organ/organism
- First purely *in silico* developed drug comes to market
- Personalised *in situ* molecular-computer 'smart drug'
- Verifiable global ecosystem models
- Verifiable global epidemic models
- Predictive model of effects of human activities on Earth's life support systems
- Understanding of Earth's life support systems
- Understanding complex biological systems, from cells and organisms to ecosystems
- Comprehensive codification of biological knowledge
- Predictive models (mathematically precise, definitional, verifiable) of biological systems (cells, organs, ecosystems)
- Understanding the make-up of the Universe

Scientific Challenges

- Initial metagenomics theory and data required for theoretical underpinning of biodiversity
- New representations of biological systems as:
  Logical schemas (ontologies)
  Programs (FORET)
  Equations (e.g. Lotka-Volterra)
- Integration of, and access to, oceanographic data, climate data, remote sensing data, taxonomic data, demographic data, disease surveillance data, species data, metagenomics data
- Population dynamic-models emerging: non-local interactions, spatially explicit, non-linear, stochastic
- Statistical and correlative models and analysis. Able to deal with sparse / noisy data; niche modelling (e.g. GLIM, GAM, GARP); pattern analysis classification; cluster analysis; novel visualisation; parameter estimation
- Understanding / codification of metabolic pathways and gene regulation network
- Main disease pathways and gene networks identified
- Computational model of carcinogenesis
- Compositional virtual cell
- Autonomous observation and analysis of complex ecological data
- 'Postdictive' modelling possible for wide range of systems
- *In silico* models in biology now in widespread use
- Cell cycle model
- Cell differentiation model
- Biological systems abstraction, perturbation and falsification / verification
- Molecular computers increasingly used to design alternative 'smart' medical therapies
- A computational theory of synthetic biology
- Biological products designed by simulation / synthetic biology

Concepts/Tools from Computer Science

- Molecular Computer proof of concept for 'smart drugs'
- Integration of computing, computer science and mathematics leads to integration of modelling, experimental and theoretical approaches in science
- Emergence of molecular machines
- Emergence of Synthetic Biology (e.g. Biobricks)
- Mobile Ambients, Brane calculus
- Stochastic π-calculus
- Statecharts and visual formalisms being used to research chemical and biological challenges
- Inductive Logic Programming used to discover families of protein folds in proteomics
- Bayesian networks used to model effects of toxins in metabolic pathways
- Application of Kolmogorov complexity concepts start to enable understanding and measurement of biological complexity across scales
- Codification of Biology begins: Emergence of Abstract Machines for biology (e.g. using Process algebra [π-calculus – SPiM, PEPA])
- Multi-scale modelling: micro-macro model integration; stochastic/deterministic model integration; hierarchical modelling architecture
- Active Learning techniques proliferating in science – towards autonomous experimentation (and understanding the brain)
- Integration of sensors (environmental, physiological, chemical) and machine learning and data management – towards autonomous experimentation
- Scaling of Bayesian inference to very large data sets
- Mathematical and empirical verification / proofs of computational models
- Autonomous Experimentation – 'Artificial scientist'
- Efficient software for parallel linear and non-linear algebra applicable to full matrices of dimension – 10^4-10^5
- *In silico* – *in vivo* integration: Chemical Universal Turing Machine. A new kind of lab-on-a-chip enabling high speed experimentation, hypothesis formulation and concept formation
- Applications of algebraic concurrency theory enabling significant breakthroughs in understanding and predicting behaviour of complex natural systems
- Widespread *de novo* creation of models, theories and solutions (e.g. protein-drug candidates) from data using advanced Machine Learning
- Abstraction, descriptive complexity and formal verification concepts from computer science enable both postdictive and predictive multi-scale modelling
- Autonomous experimentation and sophisticated sensor networks being used routinely to undertake otherwise impossible experiments
- Computer science concepts now core discipline in natural sciences (as mathematics are today in physical sciences)
- Compositionality enables prediction of emergent behaviours of composite entities
- Abstract Machines / codification concepts and tool evolution now start to enable initial predictive modelling from molecular to population biology level (e.g. metagenomics, species-biodiversity, epidemiological level) including non-local interactions with climate and intervention data and models
- Novel, biologically inspired, computing architectures and paradigms ("Building blocks of next century of computing")

Computational platforms

- Proof of concept of autonomous experimentation
- Computational science 'Workflow' engines (e.g. SEEK, Taverna)
- Deployment of 'Grid' infrastructures for doing 'large scale' science, e.g. LHC (CERN) and associated data acquisition and analysis systems (ROOT)
- XML and Web Services
- Metadata
- Data provenance
- Data security
- Data semantics
- Commodity HPC clusters increasingly available/affordable, ushering move towards use of industry standard/common operating systems for 'scientific computing'
- Tools for designing functional ensembles of molecules
- Geographic Information Systems (e.g. Google Earth, Virtual Earth, Terraserver) increasingly evolving and increasingly used in science (e.g. climatology, biodiversity informatics, epidemiology)
- Evolving Semantic Web
- Next generation peer-to-peer infrastructure for connecting distributed components to 'do' science
- Information retrieval and search for novel data types
- Data and metadata availability / requirement from publishers, together with increasing online availability, starts to transform publishing model
- 'Smart Data Cubes' from Online Analytical Processing (OLAP) used in science for large-scale complex data analysis and visualisation
- Expanding use of COTS (customised off the shelf) software / technology: moving away from 'build your own' in science
- Radical changes underway towards parallel and distributed software architectures / programming models as a result of mass move to multi-core processors
- Metadata published with data to make it usable
- Move towards program execution for scientific applications hosted in database (taking the application to the data, rather than the data to the application)
- Component-based programming models and software services in science emerging to exploit multi-core hardware architectures
- Large-scale sensor networks starting to proliferate
- Integration of data,metadata, open access and publishing continues to transform scientific publishing model
- 'Active XML' for ubiquitous data
- Data 'behind' scientific papers available in machine readable formats
- Tagging of each datum with context data – birth of a 'serendipity engine' after 'search engine' or 'Google labs'
- Innovative simulation and visualisation (at intersection of games technology and machine learning) enabling new insights into complex phenomena and models
- Journal / database hybrids
- Knowledge base for molecular mechanisms and architectures
- Managed platforms and 'Service-Oriented Architectures' being widely created and adopted in science and scientific computing, but also for the 'new kinds' of conceptual and technological tools in science
- Text mining & Natural Language Processing enable documents to be categorised and stored in a multidimensional "Ontosemantic File Systems for Science" (OSFS)
- Truly smart lab-books
- Development of equivalent of 'Office' for Scientists – Integrated suite of easy to use, standardised, well maintained applications spanning mathematical libraries, algorithms, data management, scientific paper writing, visualisation and theoretical tools (e.g. symbolic maths) that work together well
- Concept of intelligent interaction starts to become a reality (combining data cubes, transformations in science publishing, visualisation, distributed data management). Easy to program, easy to use interfaces and smart lab-book devices/interfaces
- Symbolic computation integrated into scientific databases and programming languages
- Advances in database technology starts to enable: Scalability, distributed optimisation, fault-tolerance, self-management and integrations of heterogeneity, metadata and schematisation.
- Functionalised molecular building blocks: repertoire of macromolecules with known, consistent and compositionally exploitable behaviour
- Broadly available domain-specific computational frameworks, components, reference architectures
- Ability to rapidly build complex distributed systems from shared components that are: trustworthy, predictable, reliable, scalable and extensible
- Made for purpose 'informed matter' (from supra-molecular chemistry)

Issues

- Moore's law re-defined: move to parallel, multi-core processors – and consequent need to define architectures for multi-core computing
- Emergence of communities that build / share scientific / statistical computing tools (e.g. 'R' statistical computing and graphics project / community)
- Data curation: how, who and who pays?
- Need for better collaboration between computer scientists and other scientists, and industry – how to build new kinds of communities
- Education – how to produce 'new kinds' of scientists now urgently needed (computationally and mathematically highly literate)
- Commercial software enterprises increasingly working with science community to freely provide code, applications and components to science community on shared-source basis
- How to ensure new kinds of tools are user (scientist) friendly?
- 'Data deluge' continues to increase
- Need to raise public awareness of science and global challenges
- New kinds of communities emerging as a result of need and co-involvement of governments, science community and industry
- Open Source development, investment and economic model starts to mature
- Challenge of sustaining / re-enforcing the scientific principle under increasing complexity and computationalisation: falsification, repeatability, public domain, competing schools of thought
- Versioning of datasets, models and algorithms used routinely and widespread in science
- Education – how to produce 'new kinds' of scientists now urgently needed (computationally and mathematically highly literate)
- Innovation in computational science platforms, software frameworks and applications flowing back into, and radicalising, mainstream enterprise / business computing
- Overwhelming software complexity demands 'industrial-scale' and industry-wide support to ensure reliability, robustness
- Education – how to produce 'new kinds' of scientists now urgently needed (computationally and mathematically highly literate)

2005 | 2010 | 2015 | 2020

Notes:

This draft roadmap is intended to be used as a simple guide to some of the developments underway or required in science and computing towards 2020.

The items in the roadmap represent a selection of the key milestones and challenges either underway, expected or required. We have not, however, attempted to incorporate every development possible or required. That is, our draft roadmap does not represent an exhaustive set of items.

In order to keep this draft roadmap simple, we have focused mainly on the biological domain –but developments in other research areas (such as energy and particle physics) will face similar challenges and be underpinned by similar milestones.

The linkages (dependencies, relationships) between elements (key milestones or challenges) do not appear on the roadmap because of the extent and complexity of such linkages. In order to keep the roadmap visually clean and readable, even the primary linkages have been omitted.